

Aditya Kulkarni

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SUMMARY

Platform and backend engineer with production Python, cloud (AWS/GCP), observability, and ML data pipeline experience. Built Kafka-driven indexing pipelines and Prometheus/Grafana/OpenTelemetry stacks; deployed on Google Cloud Run; ran Spark and Kafka at Capgemini (15% latency reduction). Shipped 10+ features across 7 production deployments at Isomer AI (394 commits, Docker, CI/CD). AlgoPulse: 2,000 async req/sec via Temporal.io + Kafka. MS CS NC State (May 2026).

EXPERIENCE

- Product Engineer Intern

Isomer AI

May 2025 - Present

 - Shipped 10+ backend features across 7 production deployments in TypeScript, Node.js, Express, PostgreSQL, Redis, and Docker; built microservices with CI/CD via GitHub Actions, Sentry observability, Auth0 authentication, and Temporal.io workflow orchestration -- 394 commits.
 - Designed REST API contracts and database schemas end-to-end; built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage) and RBAC.
- Graduate Research Assistant

NC State University

Apr 2025 - Present

 - Designed and deployed a cloud-native genomic data platform on Google Cloud Run serving 500GB of datasets to 100+ researchers worldwide; improved ML pipeline accuracy by 12% via hyperparameter tuning and time-series analysis.
- Software Developer Intern

Mphasis (HP Company)

Jun 2024 - Aug 2024

 - Automated CI/CD deployments on AWS and GCP, integrated Grafana monitoring dashboards, and built an ML-powered health claim processing tool using OCR, NLP, and classification models improving accuracy by 25%.
- AI/ML Intern

Capgemini

Jun 2023 - Aug 2023

 - Built distributed ML data pipelines using Hadoop and Apache Spark for large-scale batch processing and Apache Kafka streaming for real-time ingestion, reducing processing time by 15% and end-to-end latency by 20%.

PROJECTS

- VectorLift -- ML Search Pipeline

Pipelines / Observability

Production-grade ML indexing pipeline: Kafka-driven event-based corpus ingestion into Elasticsearch and Qdrant vector store; BM25 baseline, dense bi-encoder retrieval (sentence-transformers), and cross-encoder re-ranker; FastAPI serving with per-stage latency tracing; Prometheus metrics and Grafana dashboards; NDCG@10/MRR@10 evaluation across 6 configurations.

Python, PyTorch, Kafka, Elasticsearch, Qdrant, FastAPI, Prometheus, Grafana, Docker
- PulseRank -- Distributed Systems

Scalability / Observability

Fault-tolerant CQRS event-driven platform: MySQL outbox + idempotency keys for exactly-once delivery, MongoDB TTL projections, Redis sorted sets for sub-ms reads; SSE live push; k6 load benchmarks (10K concurrent users); OpenTelemetry tracing, Prometheus metrics, Grafana dashboards; Testcontainers concurrency tests.

TypeScript, NestJS, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus, Grafana
- AlgoPulse -- Scalable Orchestration

Cloud / Scaling

Re-architected from API polling to event-driven streaming after diagnosing throughput bottleneck under load; Temporal.io durable workflow state with fault-tolerant retries, activity checkpointing, and saga-pattern rollback; Apache Kafka for partitioned worker coordination. Reached 2,000 async LLM inference req/sec with zero message loss.

TypeScript, Node.js, Supabase Edge Functions, Temporal.io, Kafka, Docker
- Inventory Reservation Service

Backend / Production

Transactional inventory API in Java 21 + Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations; idempotent POST endpoints with RFC 9457 errors and automated expiry via scheduled cleanup. Concurrency verified with Testcontainers. Flyway migrations, Spring Security RBAC, multi-stage Docker build, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Testcontainers, Docker, GitHub Actions

EDUCATION

- North Carolina State University -- MS Computer Science

Aug 2024 - May 2026

Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms
- Mumbai University -- BTech Computer Science | CGPA: 9.32/10

Aug 2020 - May 2024

Machine Learning, Big Data Infrastructure, DS & Algorithms, Blockchain

SKILLS & AWARDS

Languages: Python, TypeScript, Java, JavaScript, SQL, C | **Cloud / Infra:** AWS, GCP, Docker, Kubernetes, Terraform, CI/CD | **Data / Pipelines:** Kafka, Spark, Hadoop, Temporal.io, Elasticsearch, Qdrant | **Observability:** Prometheus, Grafana, OpenTelemetry, Sentry | **Backend:** Node.js, FastAPI, Spring Boot, Express, PostgreSQL, MySQL, MongoDB, Redis

HackNC24 Top 10 (100+ teams, 48h) | **Sanjeevani Top 1%** (12 of 1,100+) | **IEEE Published** -- Mapify (500+ attendees) | **Google DSC Lead** (500+ students)